SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

Name	**Dinnepati Sindhu Prasad**
Register Number	**192311271**

COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct
I Dinnepati Sindhu Prasad (192311271) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Dinnepati Sindhu Prasad

Assessment Tasks Short Answer Test
1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected subpoints completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

A) Cloud computing is a technology that provides computing services like servers, storage, databases, networking, and software over the internet. Instead of buying and maintaining physical hardware, users can access these services whenever they need and pay only for what they use.

Characteristics:

- On-demand self-service: Users can create or remove computing resources whenever they need without contacting the service provider.
- Broad network access: Cloud services can be accessed through the internet using devices like laptops, mobiles, and tablets.
- Resource pooling: The cloud provider shares resources among many users while keeping each user's data secure.

2.Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

A) Cloud computing has developed through several stages over time.

- First organizations used mainframe computers, where many users shared one powerful computer.
- Then came personal computers, allowing individuals to work on their own systems.
- After that client-server computing connected computers through networks to share resources.
- The introduction of virtualization made it possible to run multiple virtual machines on one physical server, improving efficiency.
- Finally with the growth of the internet and web applications, software became available online, leading to modern cloud computing platforms that provide services on demand.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

A) Server virtualization is an important technology in cloud computing. It divides one physical server into multiple virtual servers, allowing different operating systems and applications to run independently on the same hardware.

Difference between Virtualization and Cloud Computing:

- Virtualization is a technology that creates virtual machines from a single physical server.
- Cloud computing is a service model that delivers computing resources over the internet.
- Virtualization focuses on efficient use of hardware resources.
- Cloud computing focuses on providing services like storage, servers, and software to users.
- Virtualization is the foundation that makes cloud computing possible.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications. A).

SOAP (Simple Object Access Protocol):

- SOAP is a protocol used for web services.
- It mainly uses XML for exchanging data.
- It is more secure and follows strict standards.
- It requires more bandwidth.

REST (Representational State Transfer):

- REST is an architectural style for web services.
- It supports multiple data formats such as JSON, XML, and HTML.
- It is lightweight, simple, and faster than SOAP.
- It uses less bandwidth and is easier to develop.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

A).

- IaaS (Infrastructure as a Service): It provides virtual servers, storage, and networking over the internet. Users manage the operating system and applications. Example: Amazon EC2.
- PaaS (Platform as a Service): It provides a platform where developers can build, test, and deploy applications without worrying about hardware management. Example: Google App Engine.
- SaaS (Software as a Service): It provides ready-to-use software over the internet. Users can access it through a web browser without installing it. Example: Google Docs.
- XaaS (Anything as a Service): It is a broad term that includes any IT service provided through the cloud, such as storage, security, or databases. Example: Microsoft 365.